Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The vapour pressure of a solution of the liquids $\mathrm{A}\left(\mathrm{p}^{\circ}=80\ \mathrm{mmHg}\right.$ and $\left.\mathrm{x}_{\mathrm{A}}=0.4\right)$ and $\mathrm{B}\left(\mathrm{p}^{\circ}=120\ \mathrm{mmHg}\right.$ and $\left.\mathrm{x}_{\mathrm{B}}=0.6\right)$ is found to be $100\ \mathrm{mmHg}$. It shows that the solution exhibits

(A) positive deviation from ideal behaviour

(B) negative deviation from ideal behaviour

(C) ideal behaviour

(D) positive deviation for lower conc. and negative for higher conc.

Q.2

A solution containing components A and B follows Raoult's law when

(A) $\mathrm{A}-\mathrm{B}$ attraction force is greater than $\mathrm{A}-\mathrm{A}$ and $\mathrm{B}-\mathrm{B}$

(B) $\mathrm{A}-\mathrm{B}$ attraction force is less than $\mathrm{A}-\mathrm{A}$ and $\mathrm{B}-\mathrm{B}$

(C) $\mathrm{A}-\mathrm{B}$ attraction force remains same as $\mathrm{A}-\mathrm{A}$ and $\mathrm{B}-\mathrm{B}$

(D) Volume of solution is different from sum of volume of solute and solvent

Q.3

Two 1-litre flask A and B are connected to each other by a valve which is closed. Flask A has benzene in equilibrium with its vapours at $30^{\circ}\mathrm{C}$. The flask B, is evacuated, and the valve is opened. Which of the following is true. If temperature is kept constant.'

(A) Some of the benzene molecules would move to flask B from flask A.

(B) Vapour pressure will be half the initial value.

(C) The vapour pressure remains unchanged

(D) Some more of the liquid benzene in flask A would evaporate.

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Q.4

For a solution of two liquids A and B it was proved that $\mathrm{P_S = x_A\left(p^{\circ}_A - p^{\circ}B\right) + p^{\circ}_B}$. The resulting solution will be

(A) Non -ideal

(B) ideal

(C) semi-ideal

(D) None of these

Q.5

All form ideal solution except

(A) $\mathrm{C_6H_6}$ and $\mathrm{C_6H_5CH_3}$

(B) $\mathrm{C_2H_6}$ and $\mathrm{C_2H_5I}$

(C) $\mathrm{C_6H_5Cl}$ and $\mathrm{C_6H_5Br}$

(D) $\mathrm{C_2H_5}$ I and $\mathrm{C_2H_5OH}$.

Q.6

Vapour pressure (in torr) of an ideal solution of two liquids A and B is given by : $P = 52X_A + 114$ where X_A is the mole fraction of A in the mixture. The vapour pressure (in torr) of equimolar mixture of the two liquids will be :

(A) 166

(B) 83

(C) 140

(D) 280

Q.7

The vapour pressure (at the standard boiling point of water) of an aqueous solution containing 28% by mass of a non-volatile normal solute (molecular mass $= 28$) will be

(A) 152 torr

(B) 608 torr

(C) 760 torr

(D) 547 torr

Q.8

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At a particular temperature, the vapour pressures of two liquids A and B are respectively 120 and $180 \mathrm{~mm}$ of mercury. If 2 moles of A and 3 moles of B are mixed to form an ideal solution, the vapour pressure of the solution at the same temperature will be (in mm of mercury)

(A) 156

(B) 145

(C) 150

(D) 108

Q.9

Which one of the following binary mixtures forms an azeotrope with minimum boiling point type?

(A) acetone-ethanol

(B) $\mathrm{H_2O - HNO_3}$

(C) benzene-toluene

(D) n -hexane- n -heptane

Q.10

The vapour pressure of pure benzene and toluene at a particular temperature are $100 \mathrm{~mm}$ and $50 \mathrm{~mm}$ respectively. Then the mole fraction of benzene in vapour phase in contact with equimolar solution of benzene and toluene is

(A) 0.67

(B) 0.75

(C) 0.33

(D) 0.50

Q.11

At $80^{\circ} \mathrm{C}$, the vapour pressure of pure liquid ' $\mathrm{A}^{\prime}$ is $520 \mathrm{~mmHg}$ and that of pure liquid ' B ' is $1000 \mathrm{~mmHg}$. If a mixture solution of ' A ' and ' B ' boils at $80^{\circ} \mathrm{C}$ and $1 \mathrm{~atm}$ pressure, the amount of ' A ' in the mixture is $(1 \mathrm{~atm}=760 \mathrm{~mmHg})$

(A) $52 \mathrm{~mol}$ percent

(B) $34 \mathrm{~mol}$ percent

(C) $48 \mathrm{~mol}$ percent

(D) $50 \mathrm{~mol}$ percent

Q.12

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A solution at $20^{\circ}\mathrm{C}$ is composed of $1.5\ \mathrm{mol}$ of benzene and $3.5\ \mathrm{mol}$ of toluene. If the vapour pressure of pure benzene and pure toluene at this temperature are 74.7 torr and 22.3 torr, respectively, then the total vapour pressure of the solution and the benzene mole fraction in equilibrium with it will be, respectively :

(A) 35.8 torr and 0.280

(B) 38.0 torr and 0.589

(C) 30.5 torr and 0.389

(D) 30.5 torr and 0.480

Q.13

Choose the correct statement with respect to the vapour pressure of a liquid among the following:

(A) Increases linearly with increasing temperature

(B) Increases non-linearly with increasing temperature

(C) Decreases linearly with increasing temperature

(D) Decreases non-linearly with increasing temperature

Q.14

For an ideal solution of two components A and B, which of the following is true?

(A) $\Delta \mathrm{H}_{\text{mixing}} < 0$ (zero)

(B) $\Delta \mathrm{H}^{*}_{\text{mixing}} > 0$ (zero)

(C) $A - B$ interaction is stronger than $A - A$ and $\mathrm{B} - \mathrm{B}$ interactions

(D) $\mathrm{A} - \mathrm{A}, \mathrm{B} - \mathrm{B}$ and $\mathrm{A} - \mathrm{B}$ interactions are identical

Q.15

Henry's law constant of oxygen is 1.4×10^{-3} mol. lit $^{-1}.\mathrm{atm}^{-1}$ at $298\ \mathrm{K}$. How much of oxygen is dissolved in $100\ \mathrm{mL}$ at $298\ \mathrm{K}$
when the partial pressure of oxygen is $0.5\ \mathrm{atm}$?

(A) $1.4\ \mathrm{g}$

(B) $3.2\ \mathrm{g}$

(C) $22.4\mathrm{mg}$

(D) $2.24\mathrm{mg}$

Q.16

Which one of the following gases has the lowest value of Henry's law constant?

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(A) N_2

(B) He

(C) H_2

(D) CO_2

Q.17

Which of the following statement is correct if the intermolecular forces in liquids A, B and C are in the order $A < B < C$?

(A) B evaporates more readily than A

(B) B evaporates less readily than C

(C) A and B evaporates at the same rate

(D) A evaporates more readily than C

Q.18

At $15^{\circ}C$ and 1 atmosphere partial pressure of hydrogen, $20\ mL$ of hydrogen measured at STP dissolves in $1\ L$ of water. If water at $15^{\circ}C$ is exposed to a gaseous mixture having a total pressure of $1500\ mm$ of Hg (excluding the vapour pressure of water) and containing 80% hydrogen by volumne, then the volume of hydrogen measured at STP that will dissolve in 1 L of water is

(A) $20.0\ mL$

(B) $31.6\ mL$

(C) $36.1\ mL$

(D) $26.3\ mL$

Q.19

The total vapour pressure of a 4 mole $\%$ solution of NH_3 in water at $293\ K$ is 50.0 torr. The vapour pressure of pure water is 17.0 torr at this temperature. Applying Henry's and Raoult's laws, the total vapour pressure for a 5 mole $\%$ solution is

(A) 58.25 torr

(B) 33 torr

(C) 42.1 torr

(D) 52.25 torr

Q.20

2 moles of liquid A $\left(P_A^o = 100 torr\right)$ and 3 moles of liquid B $\left(P_B^o = 150 torr\right)$ form a solution having vapour pressure of 120 torr. Based upon this observation one can conclude :-

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(A) Interaction between the molecules is greater than those between unlike molecules

(B) Interaction between like molecules is less than those between unlike molecules

(C) Interaction between like molecules is equal to those between unlike molecules

(D) $\Delta S_{mixing} = 0$

Answers & Solutions

Q.1 Answer:

negative deviation from ideal behaviour

Solution:

$P_{total} = p_A^o \times x_A + p_B^o \times x_B$

$= 80.0 \times 0.4 + 120.0 \times 0.6 = 104 \text{ mmHg}$

The observed P_{total} is 100 mmHg which is less than 104 mm Hg. Hence the solution shows negative deviation.

Q.2 Answer:

$A - B$ attraction force remains same as $A - A$ and $B - B$

Solution:

These two components A and B follows the condition of Raoult's law if the force of attraction between A and B is equal to the force of attraction between A and A or B and B.

Q.3 Answer:

The vapour pressure remains unchanged

Solution:

There is no change in vapour pressure.

Q.4 Answer:

ideal

Solution:

$p_s = x_A (p_A^\circ - p_B^\circ) + p_B^\circ$

$p_s = p_A^o \times x_A - p_B^o \times x_A + p_B^o$

$p_s = p_A^o \times x_A - p_B^o (1 - x_B) + p_B^o$

$\therefore p_s = p_A^o \times x_A + p_B^o \times x_B$

This is condition for ideal solution

Q.5 Answer:

C_2H_5 I and C_2H_5OH.

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Solution:

C_2H_5I **and** C_2H_5 **OH form non-ideal solution.**

Q.6 Answer:

140

Solution:

Total V.P.,
$P = P_A^0 X_A + P_B^0 X_B = P_A^0 X_A + P_B^O (1 - X_A)$
$= (P_A^o - P_B^o) X_A + P_B^o$
Thus, $P_B^\circ = 114$ **torr** ; $P_A^o - P_B^o = 52$
or $P_A^o = 166$ **torr**
Hence $P = 166 \times \frac{1}{2} + 114 \times \frac{1}{2} = 140$ **torr**

Q.7 Answer:

608 torr

Solution:

Moles of solute $= \frac{28}{28} = 1$;
moles of water $= \frac{100-28}{18} = 4$
V.P. of solution
$= P_{H_2O}^o \times X_{H_2O} = 760 \times \frac{4}{5} = 608$ **torr**

Q.8 Answer:

156

Solution:

Vapour pressure of solution $= P_A + P_B$
$= P_A^o X_A + P_B^O X_B = \frac{120 \times 2}{5} + \frac{180 \times 3}{5}$
$= 156$ mmHg

Q.9 Answer:

acetone-ethanol

Solution:

Minimum boiling azeotrope is formed by solution showing positive deviation. e.g. acetone − ethanol.

Q.10 Answer:

0.67

Solution:

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Total vapour pressure = vapour pressure of pure benzene + vapour pressure of toluene

$= 100 + 50 = 150$ mm

We know, $P^{\circ}_{C_6H_6} = P \times X_{C6H6}$

$100 = 150 \times X_{C6H6}$

$X_{C6H6} = \frac{100}{150} = 0.67$

Q.11 Answer:

50 mol percent

Solution:

At 1 atmospheric pressure the boiling point of mixture is $80^{\circ}C$.

At boiling point the vapour pressure of mixture, $P_T = 1$ **atmosphere** $= 760$ mmHg. **Using the relation,**

$P_T = P^{\circ}_A x_A + P^{\circ}_B x_B$, **we get**

$P_T = 520x_A + 1000\,(1 - x_A)$

$\{\because P^{\circ}_A = 520$ mmHg

$P^{\circ}_B = 1000$ mmHg; $x_A + x_B = 1$

or $760 = 520x_A + 1000 - 1000x_A$ **or** $480x_A = 240$

or $x_A = \frac{240}{480} == \frac{1}{2}$ **or 50 mol percent**

Q.12 Answer:

38.0 torr and 0.589

Solution:

Total V.P. of solution $= P^{\circ}_A x_A + P^{\circ}_B x_B$

Given, $P^{\circ}_A = 74.7$ **torr,** $P^{\circ}_B = 22.3$ **torr**

$n_{benzene} = 1.5$ mol, $n_{toluene} = 3.5$ mol

$n_{solution} = 1.5 + 3.5 = 5$ mol

$x_A = \frac{n_{benzene}}{n_{solution}} = \frac{1.5}{5}$

Total V.P. of solution

$= \left(\frac{1.5}{5} \times 74.7 + \frac{3.5}{5} \times 22.3\right)$ **torr**

$= (22.4 + 15.6)$ **torr** $= 38$ **torr**

Mole fraction of benzene in vapour form $=$

$\frac{22.4}{38} = 0.589$

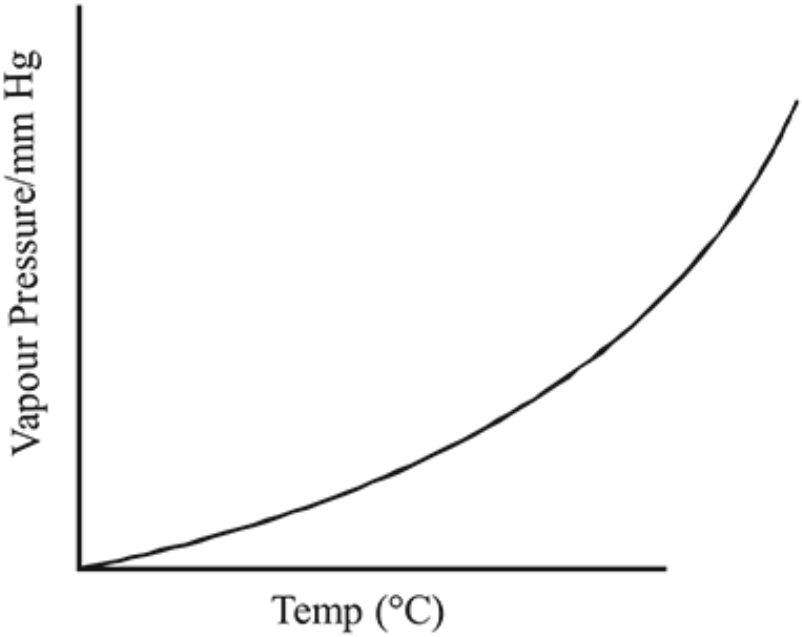
Q.13 Answer:

Increases non-linearly with increasing temperature

Solution:

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If we plot a graph between the vapour pressure and the temperature. We would get a curve that rises faster as T increases, giving a curved line.



Q.14 Answer:

$A - A, B - B$ and $A - B$ interactions are identical

Solution:

Solutions in which solute - solute and solvent-solvent interactions are almost similar to solute solvent interactions are known as ideal solution.

Q.15 Answer:

$2.24 mg$

Solution:

According to Henry's law, $m = k \times p$

given $K_H = 1.4 \times 10^{-3}$

$p_{O_2} = 0.5 or$

$p_{O_2} = K_H \times x_{O_2}$

$\therefore x_{O_2} = \frac{0.5}{1.4 \times 10^{-3}}$

No. of moles; $n = \frac{m}{M}$

$0.7 \times 10^{-4} = \frac{m}{32}$

$m = 22.4 \times 10^{-4} g = 2.24 mg$

Q.16 Answer:

CO_2

Solution:

According to Henry's law the mass of a gas dissolved per unit volume of solvent is proportional to the pressure of the gas at constant temperature $m = Kp$ **i.e. as the solubility increases, value of Henry's**

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law constant decreases. Since CO_2 is most soluble in water among the given set of gases. Therefore CO_2 has the lowest value of Henry's law constant.

Q.17 Answer:

A evaporates more readily than C

Solution:

Lesser the intermolecular forces, the more the volatile character.

Q.18 Answer:

$31.6\ mL$

Solution:

From the given data $P_{H2} = 1500 \times 0.80$
$= 1200\ mm$ **of** $Hg = \frac{1200}{760}$ **atmosphere** $= 1.58$
atmosphere If 's are volumes of gas dissolved by same volume of liquid, then from Henry's law
$\frac{V_2}{V_1} = \frac{P_2}{P_1}$; **or** $\frac{V_2}{20} = \frac{1.58}{1.0}$
or $V_2 = 31.60\ mL$

Q.19 Answer:

58.25 torr

Solution:

The relationship between the relative lowering of vapour pressure of solvent and the molecular weight of solute and solvent is as shown below. $\frac{\Delta P}{P} = \frac{W}{M} \times \frac{m}{w}$
4 mole\% ammonia means 68 g **ammonia in** 1800 g **water.** $\frac{17-P}{17} = \frac{68}{17} \times \frac{18}{1800}$
Vapour pressure of water $P = 17 - 0.68 = 16.32$
Vapour pressure of ammonia $= 50.0 - 16.32 = 33.68$ **torr**
From henry's law, $S = KP$
Hence, $K = \frac{S}{P} = \frac{4}{33.68}$
5 mole\% ammonia means 85 g **ammonia in** 1800 g **water.**
$\frac{17-P}{17} = \frac{85}{17} \times \frac{18}{1800}$
Vapour pressure of water $P = 17 - 0.85 = 16.15$
From henry's law, $S = KP$
$5 = \frac{4}{33.68} \times P$
Vapour pressure of ammonia $P = 42.1$
Total vapour pressure $= 16.15 + 42.1 = 58.25$ **torr**

Q.20 Answer:

Interaction between like molecules is less than those between unlike molecules

Solution:

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For ideality,

$P = P_A^o X_A + P_B^o X_B$

$P = 100 \times \frac{2}{5} + 150 \times \frac{3}{5} = 40 + 90$

$P = 130 \quad \text{torr} > \textbf{given } P = 120 \quad \text{torr}$

$\Rightarrow$ **-ve deviation** $\therefore$ A - B **interaction** $>$ A - A & B - B **interactions**

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